

Designing a Smart VR Painting System with Multisensory Interaction for Immersive Experience

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ABSTRACT

VR painting has become increasingly popular for its potential to provide a less restricted painting environment. However, limited sensory feedback in existing VR painting systems detracts people from fully immersive experience. To better integrate multisensory interaction into the VR painting system, we invited 12 participants to a participatory design process. Based on participants' feedback, we envision the design of a smart VR painting system, which integrates visual, audio, haptic, and smell feedback to enhance immersion. Meanwhile, AI's capability is adopted to reduce the difficulty of painting in VR.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

As Virtual Reality (VR) technology advances, people are able to doodle or paint in a virtual environment (VE) without being restricted by physical limitations. Similarly to improvising [5] or learning [1] in VR, the level of immersion plays a dominant role in shaping VR painting experience. Immersion describes the degree to which a system involves people in the VE, which has been proven effective at promoting creativity [3].

However, the lack of multisensory interaction in existing VR painting systems somewhat detracts people from fully immersive painting experience. As suggested by prior work, multisensory interaction can activate people's brain and facilitate the engagement of body [4]. Taking advantage of this, existing work created memorable experiences in tourism [6] and increased the expressiveness of visual art [2]. Despite that, existing VR painting systems mainly create immersion by shaping realistic environment (e.g., Painting VR) or providing infinite 3D painting space (e.g., Tilt Brush).

Motivated by this, we attempt to incorporate multisensory interaction to the VR painting system for immersive experience. We first invited 12 participants into a participatory design, which helped us define issues related to immersion in existing VR painting applications and explore how to integrate multisensory interaction into VR painting. Our findings revealed that participants perceived the existing applications as lacking sensory feedback, with complex interactions that diminished their immersion. Accordingly, they would

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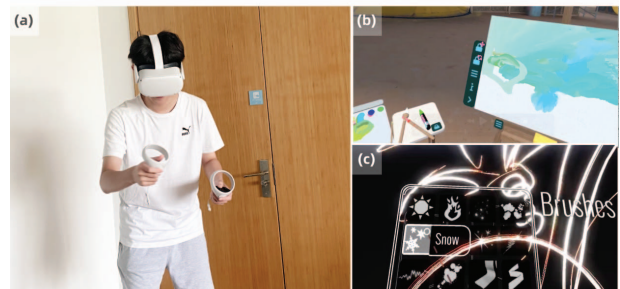


Figure 1: (a) The apparatus of participatory design. (b) The screen shot of Painting VR. (c) The screen shot of Tilt Brush.

like to interact with the virtual environment in a sensory way and receive additional painting assistance. Based on the above findings, we envision the main functionality and interaction design of the multisensory VR painting system. The system integrates multiple sensory channels including vision, audition, touch, and olfaction into a natural scene to provide interactive sensory feedback. In addition, we intend to apply artificial intelligence (AI) for generating richly textured cartoon images based on people's rough 2D sketches, helping users obtain fascinating paintings rapidly.

2 PARTICIPATORY DESIGN

2.1 Methods

To understand potential factors concerning immersion and explore how to integrate multisensory interaction into VR painting, we conducted a participatory design with 12 participants (P1-P12). Five participants received systematic training on painting, and seven participants had experience using the VR headset. Each participant was asked to wear Oculus Quest 2 to use Painting VR and Tilt Brush, respectively (Fig. 1). They were two of the most popular commercial VR painting applications with relatively complete functionality, and their efforts to create immersion were similar to ours. For each application, participants were free to use it until they completed a painting. After using, participants were asked to describe their feeling of immersion during painting with examples.

2.2 Findings

The participatory design process revealed issues about the lacked multisensory feedback and the complex interactions. For partici-

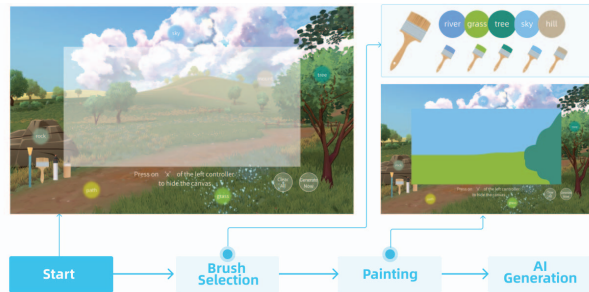


Figure 2: The painting workflow and main interfaces.

pants without VR experience, the immersion brought by the realistic environment in Painting VR surprised them at the very beginning. However, as the painting process went on, participants found that they could easily be disturbed by the noise in reality. Furthermore, P3 found that “the sensory feedback of existing VR painting applications was somewhat limited” when compared with a VR rhythm game with visual effects, music, and vibration he once played. In addition, as described by P5 and P9, the complex interactions negatively impacted immersion. For example, they did not even know how to select brushes in Painting VR without help. As for Tilt Brush, both painting lovers and expert painters encountered difficulty in locating the brush and strokes in 3D space and creating the ideal shapes. As a result, participants tended to think about the painting mechanism instead of immersing themselves in painting.

Accordingly, to facilitate immersion in VR painting systems, participants raised the expectation for sensory interaction with environment and mentioned adding external assistance to beautify their paintings. For example, P2 specifically stated that “if there were sounds and smells from the environment, I will possibly feel more immersive”. In addition, P7 advised that “when I chose the fire brush in Tilt Brush, I expected to feel the heat from the fire”.

3 DESIGN OVERVIEW

Grounded on the feedback from the participatory design, we envision the interaction workflow (see Fig.2) and main functionality of the system. As prior work proved that natural scenes could hopefully provoke creativity [1], we reproduce the scene of field in VR. The field is composed of several natural objects including tree, grass, stone, sky, hill, and path. To simplify the interaction, we allow users to choose color by pointing to a natural object. For example, if the user wants to paint a stone, he/she can point to a stone in the scene.

3.1 Multisensory Interaction

Our design of multisensory interaction was inspired by Season Traveller, which tailored scents and haptics to different natural scenes [6]. In our system, multisensory feedback is integrated throughout the painting process, with different natural objects triggering different forms of feedback. For example, when a user points to the grass, he/she will smell the odor of freshly mown grass; when a user points to a tree, he/she will hear the sound of the wind blowing, smell the wood, and feel the distinct sensation of wind.

For the technical implementation of sensory feedback, we plan to adopt more desktop devices (see Fig.3), which can reduce the physical load on users. The visual and audio feedback can be provided by the VR headset. Fans will be used to provide haptic feedback such as the sensation of wind. As for smell feedback, we seek to adopt a multi-channel odor display device to support independent control of different scents.

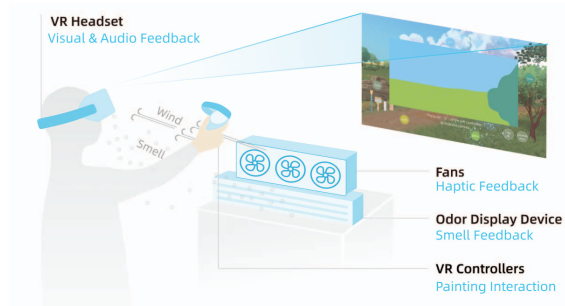


Figure 3: Overview of the interaction devices.

3.2 AI Generation

With the recent advances in AI, we attempt to provide painting assistance in VR through AI’s generation ability. Prior work made it possible to generate images from the semantic layout by adopting a conditional generative adversarial network (GAN) [7]. In our system, we aim to implement the above-mentioned AI-assisted painting process in VR. In this way, users only need to express their ideas by outlining the layout of the painting in VR. Then our system transfers users’ rough 2D sketches into the trained GAN and automatically generates richly textured cartoonish painting for users. This process helps users concentrate more on their inner thoughts and get a satisfying painting rapidly. Even users without extensive painting experience can create pleasant paintings in VR with our system.

4 CONCLUSION

We illustrate the design of a VR painting system, which integrates multisensory interaction and AI’s capability to create immersive experience. To improve this system, we will evaluate it through comprehensive user study and iterated it according to feedback. We hope our work can inspire the design of future VR painting systems.

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